

Optimization

3.2.4 Box's method — a simplified version of EVOP (evolutionary operation)

José Firmino Aguilar Madeira

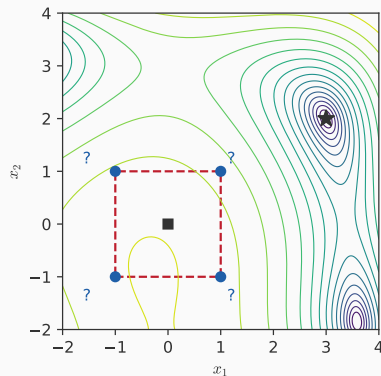
From a shape-changing simplex to a box that does not change

Nelder–Mead **adapts the shape** of the simplex using the values of f . What if we always keep the **same geometric pattern**?

In 2D, around the current point $\mathbf{x}$, the four vertices of a box:

$$\mathbf{x} + \frac{1}{2}(\pm\Delta_1, \pm\Delta_2).$$

Here $\mathbf{x} = (0, 0)$, $\Delta = (2, 2)$, on Himmelblau.



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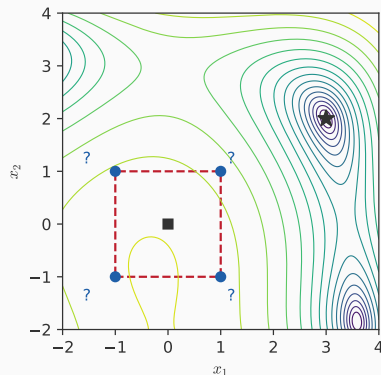
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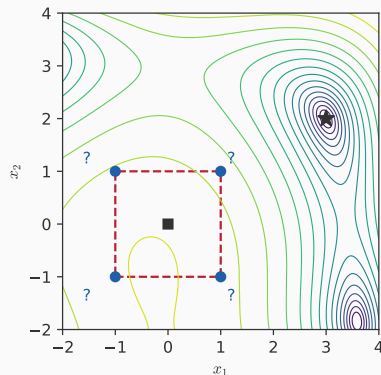
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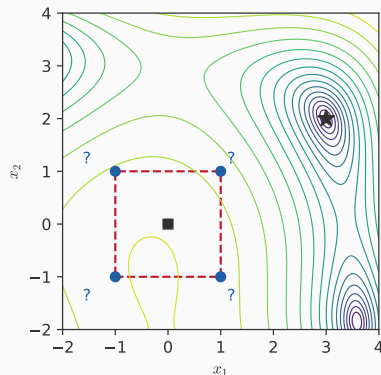
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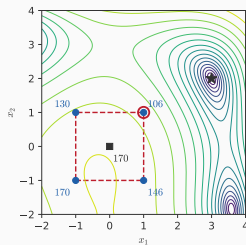
- | | |
|-----------------------|--|
| random search (3.2.1) | points without structure |
| grid (3.2.2) | fixed structure on the whole domain |
| Nelder–Mead (3.2.3) | adaptive geometry, which moves |
| Box (3.2.4) | fixed pattern, which moves |



One iteration by hand

$$f(\mathbf{x}) = (x_1^2 + x_2 - 11)^2 + (x_1 + x_2^2 - 7)^2, \quad \mathbf{x} = (0, 0),$$
$$f(\mathbf{x}) = 170, \quad \Delta = (2, 2).$$

vertex	f
$(-1, -1)$	170
$(1, -1)$	146
$(-1, 1)$	130
$(1, 1)$	106

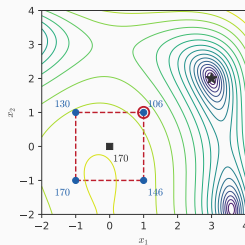


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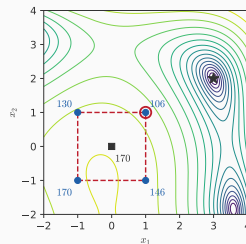
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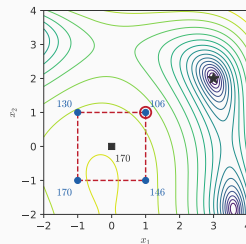
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We have just built the whole algorithm.



Origin

G. E. P. Box (1957), *Evolutionary Operation: tuning industrial processes in production* (Box worked at ICI), with small variations around the current conditions; each evaluation was a batch. The original EVOP is experimental and statistical; here we use a **simplified deterministic version** (not to be confused with the *Complex* method of M. J. Box, 1965 — another author, another algorithm).

The algorithm, and its cost

Box: starting point $\mathbf{x}_0$, sides Δ_i ,
tolerance ε_x

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2: while  $\|\Delta\| \geq \varepsilon_x$  do
3:   generate the  $2^n$  vertices  $\mathbf{x} + \frac{1}{2}(\pm\Delta_1, \dots, \pm\Delta_n)$ 
4:   evaluate  $f$  at all of them;  $\mathbf{x}' \leftarrow$  the best
5:   if  $f(\mathbf{x}') < f_x$  then  $\mathbf{x} \leftarrow \mathbf{x}'$ ;  $f_x \leftarrow f(\mathbf{x}')$   $\triangleright$  move
6:   else  $\Delta \leftarrow \Delta/2$   $\triangleright$  shrink
7:   end if
8: end while
9: return  $\mathbf{x}$ 
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- The Δ_i may differ per coordinate; stop when the box becomes smaller than ε_x .
- An implementation with a cache can reuse points already evaluated; this deck's counts always evaluate all 2^n vertices.

$$n_f = 2^n \text{ per iteration}$$

n	2	3	10	20
vertices	4	8	1 024	$\approx 10^6$

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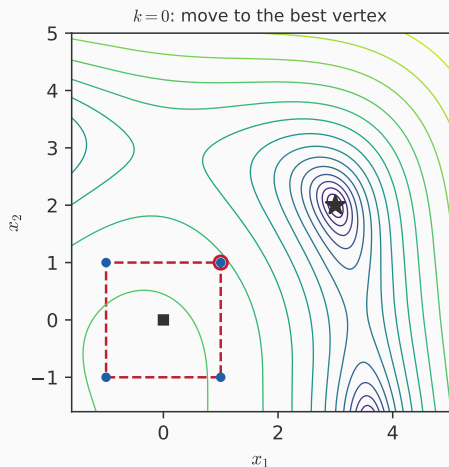
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The curse of dimensionality again — for a different reason:

- random search and grid (3.2.1–3.2.2): the **volume** to cover;
- Box: the **number of vertices** to test.

Same consequence: dimension is very costly.

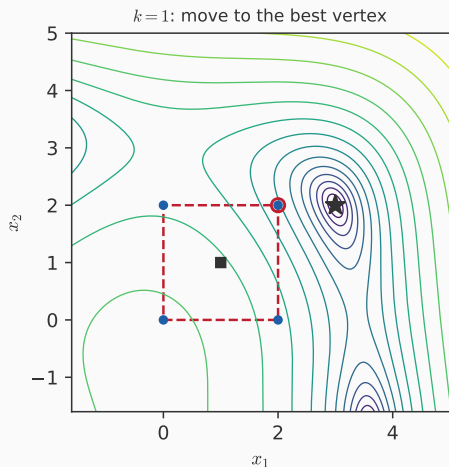
Himmelblau from (0,0): move or shrink



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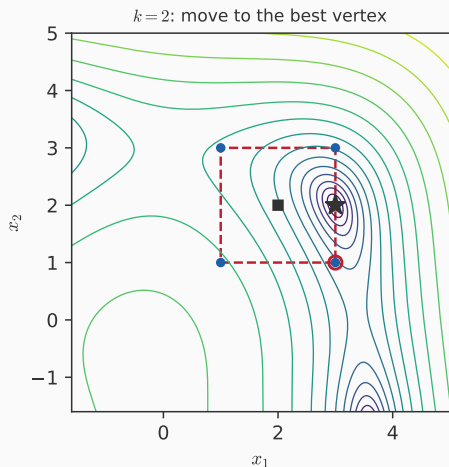
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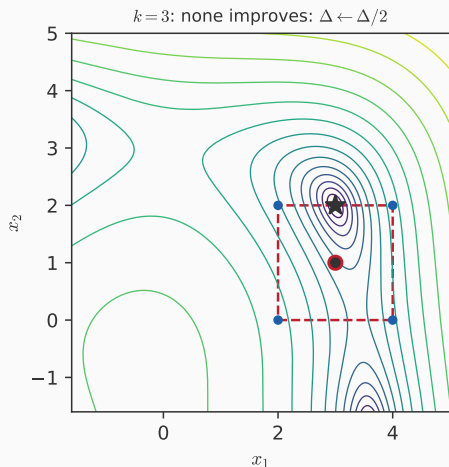
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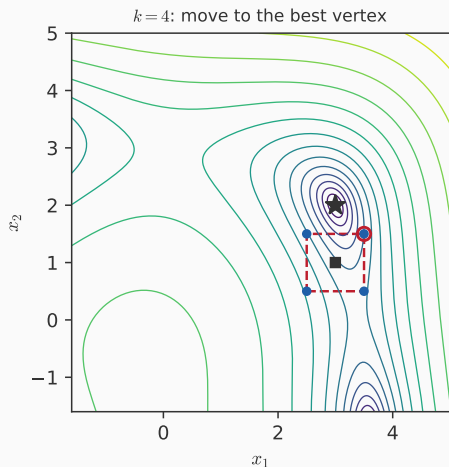
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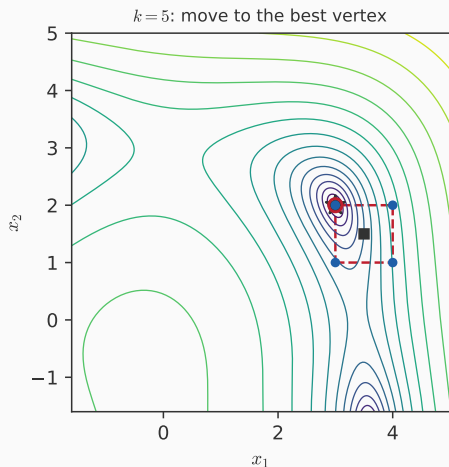
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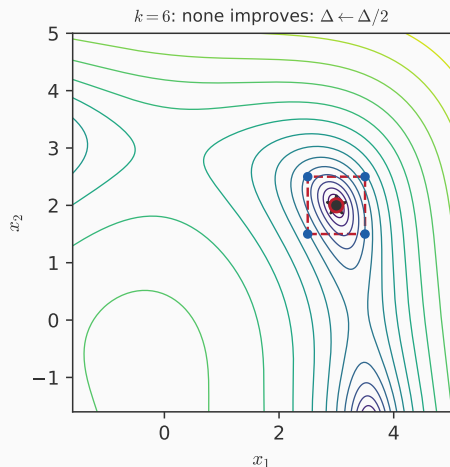
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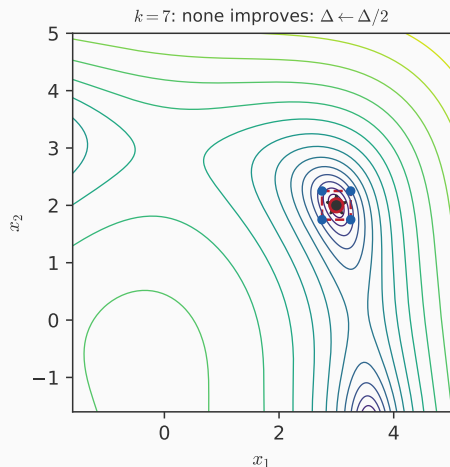
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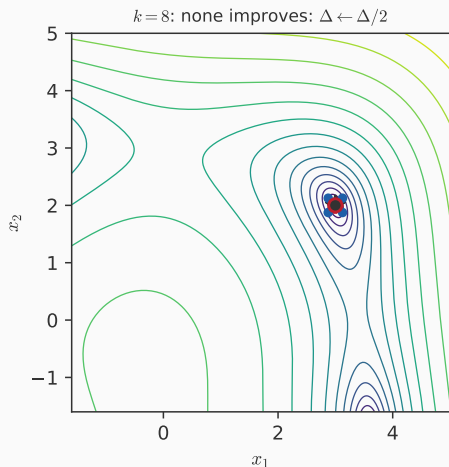
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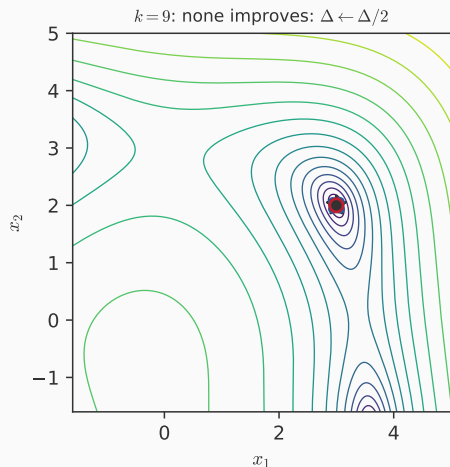
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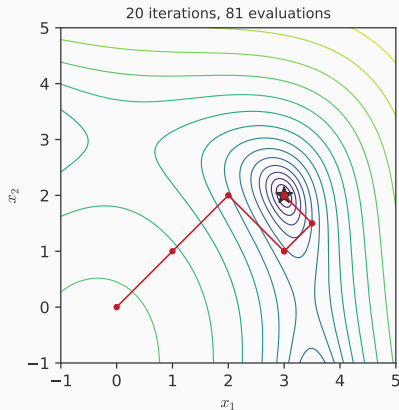


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The box **moves** with fixed size,
 $(0, 0) \rightarrow (1, 1) \rightarrow (2, 2) \rightarrow (3, 1)$; at $(3, 1)$ no vertex improves and it **shrinks** (Δ : $2 \rightarrow 1$); then
 $(3, 1) \rightarrow (3.5, 1.5) \rightarrow (3, 2)$ and it only shrinks.

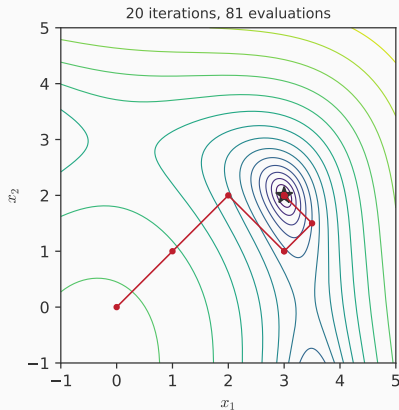
The complete path: what the box is doing



20 iterations, $n_f = 4 \times 20 + 1 = 81$ until
 $\|\Delta\| < 10^{-4}$.

- While there is improvement, the centre jumps from vertex to vertex: all coordinates change by $\pm\Delta_i/2$, so the step has fixed length $\frac{1}{2} \|\Delta\|$. Only when it stagnates does it decrease.
- It reaches (3,2) *exactly* because the minimum lies on the grid of steps $\Delta/2$.

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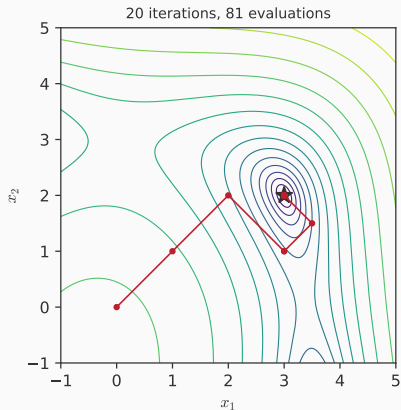


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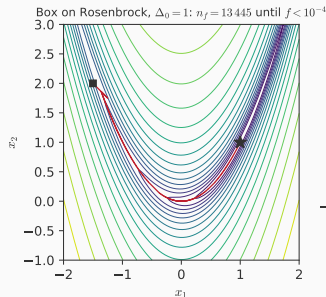
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No. We only know that, in that pattern and at that scale, no tested vertex improved. A small box does not certify stationarity at finite Δ , nor a local minimum, nor $\|\mathbf{x} - \mathbf{x}^*\| = O(\varepsilon_x)$.

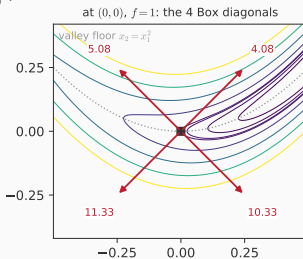
Why does it fail in curved valleys?



What is wrong with changing all coordinates at once?

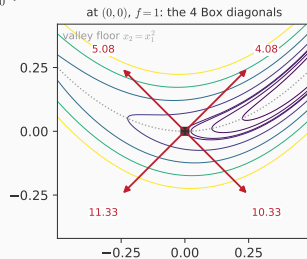
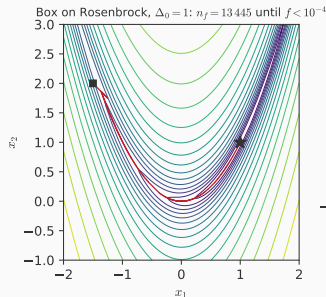
On Rosenbrock, from $(-1.5, 2)$ with $\Delta_0 = 1$, the box has to shrink from 1 to $\approx 6 \cdot 10^{-5}$ to make progress in the valley, and then it advances in tiny steps:

$n_f = 13\,445$ until $f < 10^{-4}$ (with $\varepsilon_x = 10^{-6}$).



At $(0, 0)$, $\Delta = (0.5, 0.5)$ (vertices ± 0.25): **no diagonal** ($\pm 1, \pm 1$) improves $f = 1$ (4.08 to 11.33).

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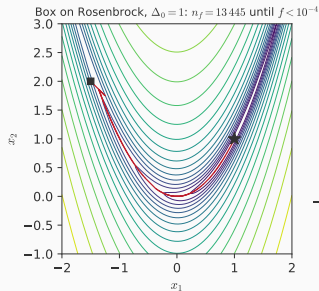
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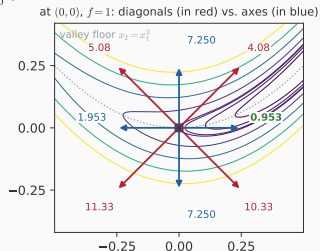
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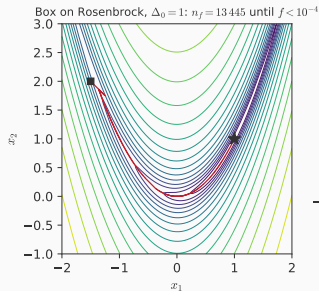


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Box concluded that there is no improvement. Is there really no descent direction?

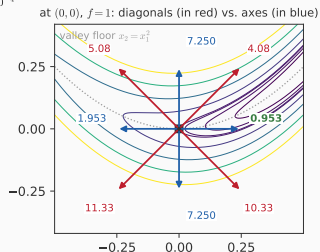
There is: the axis $+e_1$ improves, $f(0.25, 0) = 0.953$.
The valley is horizontal there; the diagonals leave it.

Why does it fail in curved valleys?



What is wrong with changing all coordinates at once?

On Rosenbrock, from $(-1.5, 2)$ with $\Delta_0 = 1$, the box has to shrink from 1 to $\approx 6 \cdot 10^{-5}$ to make progress in the valley, and then it advances in tiny steps:
 $n_f = 13\,445$ until $f < 10^{-4}$ (with $\varepsilon_x = 10^{-6}$).



At $(0,0)$, $\Delta = (0.5, 0.5)$ (vertices ± 0.25): no diagonal $(\pm 1, \pm 1)$ improves $f = 1$ (4.08 to 11.33).

Box concluded that there is no improvement. Is there really no descent direction?

There is: the axis $+e_1$ improves, $f(0.25, 0) = 0.953$.
The valley is horizontal there; the diagonals leave it.

diagonals $(\pm 1, \dots, \pm 1)$ vs. axes $\pm e_i$

The price of Box; on the computer

Advantages

order 0; deterministic; very simple; scale Δ ; controllable per variable; the evaluations of the 2^n vertices can be done in parallel.

Limitations

- 2^n evaluations per iteration;
- only the 2^n diagonals $D = \{(\pm 1, \dots, \pm 1)\}$: depends on the coordinate orientation;
- inefficient in narrow, curved valleys; local;
- small box $\nrightarrow$ stationarity (at finite Δ);
- noise can change the choice of the best vertex.

On Rosenbrock from $(-1.5, 2)$, until $f < 10^{-4}$: Box ($\Delta_0 = 1$)
 $n_f = 13\,445$; Nelder–Mead $n_f = 166$ (same problem, same starting point, same criterion; with these parameters).

The core, in MATLAB:

```
S = dec2bin(0:2^n-1) - '0';  
S = 2*S - 1; % rows of +-1  
V = x + 0.5*S.*Delta; % vertices  
Fv = arrayfun(@(i) f(V(i,:)), 1:2^n);  
[fmin, i] = min(Fv);  
if fmin < fx  
    x = V(i,:); fx = fmin; % move  
else  
    Delta = Delta/2; % shrink  
end
```

In 2D, S has the 4 rows $(-1, -1)$, $(-1, 1)$, $(1, -1)$, $(1, 1)$.

The full function (BoxEvo, with info.nfev) is in the course companion code.

KEY TAKEAWAYS — 3.2.4

1. Box uses a **box centred** at the current point and tests all 2^n diagonal combinations.
2. Improvement $\Rightarrow$ **move** to the best vertex; failure $\Rightarrow \Delta \leftarrow \Delta/2$.
3. It uses only values of f , but the cost, **2^n per iteration**, grows exponentially with the dimension: testing all diagonals is **expensive and rigid**.
4. A small box does not certify stationarity at finite Δ . For this simplified version of Box we do not present here a general stationarity guarantee.
5. NM has no general guarantee; HJ is the origin of pattern search, whose formalized versions (GPS) have stationarity in the limit under suitable assumptions (3.2.5).

Next (3.2.5): do we really need the 2^n diagonals? What if we explore **one coordinate at a time**, with only **$2n$ directions**, and then exploit the direction of progress? — the Hooke–Jeeves method.

References and resources

References Full list: *Course bibliography*, at the end of the course PDF.

- Box, G. E. P., *Evolutionary operation: a method for increasing industrial productivity*, Journal of the Royal Statistical Society, Series C (Applied Statistics) 6(2):81–101, 1957.
- Deb, K., *Optimization for Engineering Design: Algorithms and Examples*, 2nd ed., PHI Learning, 2012 — Ch. 3.

Resources

- Companion code: BoxEvo (info.nfev).